

EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.
DATE:05/03/2025	

AIM:

To develop an Online Quiz Application using HTML and Java Servlet to demonstrate handling of HTTP GET and POST methods, where GET is used to fetch quiz questions and POST is used to submit answers.

Algorithm :

1. Input:

- The user accesses the quiz page.
- The user answers multiple-choice questions.
- The form submits responses using GET and POST methods.

2. Steps for Creating and Handling the Application:

Step 1: Create an HTML Quiz Form

- Design an HTML page with:
 - Multiple-choice questions.
 - Radio buttons for answers.
- Use:
 - GET method to load or view quiz questions.
 - POST method to submit answers.

Step 2: Create the Servlet

- Create a Java Servlet class.
- Implement:
 - doGet() → to display quiz questions.
 - doPost() → to evaluate answers.

Step 3: Handle GET Request

- In doGet():
 - Display quiz questions dynamically.
 - Send HTML form as response.

Step 4: Handle POST Request

- In doPost():
 - Retrieve answers using request.getParameter().
 - Compare answers with correct ones.
 - Calculate score.

Step 5: Display Result

- Show:
 - Score
 - Correct/incorrect answers

Step 6: Deploy and Test

- Compile servlet.
- Deploy in Apache Tomcat.
- Test:
 - Loading quiz (GET)
 - Submitting quiz (POST)

Step 7: Observe Difference

- GET → displays quiz (data visible in URL).
- POST → submits answers (data hidden in request body).

CODE :

Quiz.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Quiz</title>
</head>
<body style="font-family:Segoe UI;background:#f2f6ff;">

  <h2 align="center">Online Quiz Application</h2>

  <form action="QuizServlet" method="post">

    <p>1. What is Java?</p>
    <input type="radio" name="q1" value="a"> Programming Language<br>
    <input type="radio" name="q1" value="b"> Operating System<br>

    <p>2. Which is used for web development?</p>
    <input type="radio" name="q2" value="a"> HTML<br>
    <input type="radio" name="q2" value="b"> MS Word<br>

    <p>3. Which method is secure?</p>
    <input type="radio" name="q3" value="a"> GET<br>
    <input type="radio" name="q3" value="b"> POST<br>

    <br>
    <button type="submit">Submit Quiz</button>

  </form>

</body>
</html>
```

QuizServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/QuizServlet")
public class QuizServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html><body>");
        out.println("<h2>Quiz Loaded Successfully (GET Method)</h2>");
        out.println("<a href='Quiz.html'>Start Quiz</a>");
        out.println("</body></html>");
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String q1 = request.getParameter("q1");
        String q2 = request.getParameter("q2");
        String q3 = request.getParameter("q3");

        int score = 0;

        if ("a".equals(q1)) score++;
        if ("a".equals(q2)) score++;
        if ("b".equals(q3)) score++;

        out.println("<html><body style='font-family: Segoe UI; background: #f2f6ff;'>");
        out.println("<div style='width: 400px; margin: 50px auto; background: white; padding: 20px; border-radius: 10px;'>");
        out.println("<h2>Quiz Result</h2>");
        out.println("<p>Your Score: <b>" + score + "/" + 3 + "</b></p>");
        out.println("<br><a href='Quiz.html'>Try Again</a>");
        out.println("</div></body></html>");
    }
}
```

OUTPUT :

The image displays three sequential screenshots of a web application titled "Online Quiz Application".

The first screenshot shows the application's main interface with two primary sections:

- Search Quiz (GET):** A search bar with the placeholder text "Enter quiz topic (e.g., Sports, Science)" and a blue "Search" button.
- Start Quiz (POST):** A form with input fields for "Name" (containing "Hari"), "Age" (containing "20"), a dropdown menu for "Category" (set to "General Knowledge"), and "Questions" (containing "10"). A blue "Start Quiz" button is at the bottom.

The second screenshot shows a success message:

- Quiz Submitted Successfully!** (indicated by a green checkmark icon)
- Summary details:
 - Name : Hari
 - Age : 20
 - Category : General Knowledge
 - Score : 8 / 10
- A green box with the message: "Great Job! Keep learning and improving."
- A blue "Back" link.

The third screenshot shows the search results:

- Search Results** (indicated by a magnifying glass icon)
- Text: "Showing results for: Sports"
- A table with the following data:

#	Quiz Title	Category	Questions	Action
1	Sports Quiz - Easy	Sports	10	Start Quiz
2	Sports Quiz - Medium	Sports	15	Start Quiz
3	Sports Quiz - Hard	Sports	20	Start Quiz

A blue "Back" link is located below the table.

Result :

The Notice Board interface is successfully converted into an Online Quiz Application, maintaining the same UI structure while demonstrating GET (load quiz) and POST (submit answers) functionality.